

Beyond Basic - Part 3

Written by Administrator

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This issue I will explain about the Hook HGONE which can be used to add extra commands to your Basic programs. This eliminates the need to use USR statements which can be confusing to use for people who are not familiar with machine code. It also limits the number of parameters to one.

The Hook HGONE is called by the Basic ROM before each statement is processed. This allows you to add a routine that checks if the next Basic statement is one that you have added.

For the ease of checking I will use the '[' character to identify any commands that we add.

The following routine creates two new Basic commands called [BANG and [ZAP. Each time you use these statements they will produce a sound, [BANG an explosion sound and [ZAP a laser firing sound.

These new commands can be used in your Basic programs.

e.g.

```
10 PRINT "THIS IS A LASER...":[ZAP20 FOR A=1 TO 500:NEXT A30 PRINT "THIS IS AN  
EXPLOSION...":[BANG
```

Machine Code Program

0000		1 ORIGIN		; Example of Hook HGONE
		2		;
D000		3	ORG 0D000H	
		4		;
		5 HGONE	EQU 0FF43H	; Hook Jump – Basic handler
		6		;
D000	F3	7 START	DI	
D001	210ED0	8	LD HL,CHECK	; Address to jump to

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D004	2244FF	9	LD (HGONE+1),HL	; Load into Hook
D007	3EC3	10	LD A,0C3H	; Value for JUMP
D009	3243FF	11	LD (HGONE),A	; Load into Hook
D00C	FB	12	EI	
D00D	C9	13	RET	
		14		;
D00E	FE5B	15	CHECK CP “[	; Is next statement
D010	C0	16	RET NZ	; one of our commands
D011	F3	17	DI	;
D012	E5	18	PUSH HL	; Save pointer
D013	1149D0	19	LD DE,TABLE	; Table of new commands
D016	0600	20	LD B,0	; Loop counter
D018	23	21 LOOP1	INC HL	; Increment pointer
D019	1A	22	LD A,(DE)	;
D01A	FE00	23	CP 0	;
D01C	2002	24	JR NZ,LOOP2	; End of Table?
D01E	1842	25	JR EXIT	; Yes – Back to Basic
D020	FE2E	26 LOOP2	CP “.”	; End of statement
D022	2015	27	JR NZ,LOOP3	; No keep going
D024	33	28	INC SP	; Get rid of old
D025	33	29	INC SP	; pointer
D026	E5	30	PUSH HL	; save new pointer
D027	DD2153D0	31	LD IX,JMPTBL	;
D02B	1600	32	LD D,0	;
D02D	58	33	LD E,B	;
D02E	CB23	34	SLA E	;
D030	DD19	35	ADD IX,DE	;
D032	DD6E00	36	LD L,(IX+0)	; Calculate jump
D035	DD6601	37	LD H,(IX+1)	; address
D038	E9	38	JP (HL)	;
D039	4E	39 LOOP3	LD C,(HL)	; Get next character
D03A	13	40	INC DE	; Set DE for next loop
D03B	B9	41	CP C	; Match?
D03C	28DA	42	JR Z,LOOP1	; Yes, go again
D03E	04	43	INC B	; Next statement
D03F	E1	44	POP HL	;
D040	E5	45	PUSH HL	; Restore pointer
D041	1A	46 LOOP4	LD A,(DE)	;
D042	13	47	INC DE	; Move DE to next
D043	FE2E	48	CP “.”	; statement
D045	28D1	49	JR Z,LOOP1	; Loop again
D047	18F8	50	JR LOOP4	;
		51		;
D049	42414E47	58 TABLE	DEFM “BANG.”	; Statement table
D04D	2E			
D04E	5A41592E	59	DEFM “ZAP.”	;

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D052	00	60	DEFB 0	;
		61		;
D053	74D0	62 JMPTBL	DEFW BANG	; Jump Table
D055	8CD0	63	DEFW ZAP	;
		64		;
D057	1A	65 SOUND	LD A,(DE)	; Get next value
D058	FEFF	66	CP 255	; End of List?
D05A	C8	67	RET Z	; Yes – Return
D05B	D3A0	68	OUT (0A0H),A	; Send value
D05D	13	69	INC DE	; Increment pointer
D05E	FE07	70	CP 7	; Is it register 7
D060	1A	71	LD A,(DE)	;
D061	2006	72	JR NZ,SNDLP1	; No - Continue
D063	47	73	LD B,A	;
D064	DBA2	74	IN A,(0A2H)	; Get current value
D066	E6C0	75	AND 192	;
D068	80	76	ADD A,B	; Combine values
D069	D3A1	77 SNDLP1	OUT (0A1H),A	; Send value
D06B	13	78	INC DE	; Increment pointer
D06C	18E9	79	JR SOUND	;
		80		;
D06E	E1	81 EXIT	POP HL	; Restore pointer
D06F	7E	82	LD A,(HL)	; Restore A
D070	33	83	INC SP	; Get rid of
D071	33	84	INC SP	; Basic return
D072	FB	85	EI	;
D073	C9	86	RET	;
		87		;
D074	117CD0	88 BANG	LD DE,BANG1	; Set pointer
D077	CD57D0	89	CALL SOUND	; Call sound routine
D07A	18F2	90	JR EXIT	; Exit to Basic
		91		;
D07C	09100208	92 BANG1	DEFB 9,16,2,8,3,5,7,40	
D080	03050728			
D084	06140C0A	93	DEFB 6,20,12,10,13,0,255,255	
D088	0D00FFFF			
		94		;
D08C	1194D0	95 ZAP	LD DE,ZAP1	; Set pointer
D08F	CD57D0	96	CALL SOUND	; Call sound routine
D092	18DA	97	JR EXIT	; Exit to Basic
		98		;
D094	091002C8	99 ZAP1	DEFB 9,16,2,200,3,0,12,6	
D098	03000C08			
D09C	07380D04	100	DEFB 7,56,13,4,255,255	
D0A0	FFFF			
D0A2		101 END		

Basic Loader

```
10 CLS: CLEAR 200, &HCFFF: DEFINT A-Z: A = &HD00020 READ A$: IF A$ <> "@" THEN POKE  
A, VAL("&H"+A$): A = A + 1: GOTO 20  
PRINT"
```

30

```
INSERT TAPE/DISK TO SAVE PROGRAM"  
40 PRINT"
```

```
AND PRESS ANY KEY"  
50 A$ = INPUT$(1): PRINT: PRINT "
```

```
SAVING ...."  
60 BSAVE "CHECK", &HD000, A-1  
100 DATA F3, 21, 0E, D0, 22, 44, FF, 3E, C3, 32, 43, FF, FB, C9, FE, 5B  
110 DATA C0, F3, E5, 11, 49, D0, 06, 00, 23, 1A, FE, 00, 20, 02, 18, 4E  
120 DATA FE, 2E, 20, 15, 33, 33, E5, DD, 21, 53, D0, 16, 00, 58, CB, 23  
130 DATA DD, 19, DD, 6E, 00, DD, 66, 01, E9, 4E, 13, B9, 28, DA, 04, E1  
140 DATA E5, 1A, 13, FE, 2E, 28, D1, 18, F8, 42, 41, 4E, 47, 2E, 5A, 41  
150 DATA 50, 2E, 00, 74, D0, 8C, D0, 1A, FE, FF, C8, D3, A0, 13, FE, 07  
160 DATA 1A, 20, 06, 47, DB, A2, E6, C0, 80, D3, A1, 13, 18, E9, E1, 7E  
170 DATA 33, 33, FB, C9, 11, 7C, D0, CD, 57, D0, 18, F2, 09, 10, 02, 08  
180 DATA 03, 05, 07, 28, 06, 14, 0C, 0A, 0D, 00, FF, FF, 11, 94, D0, CD  
190 DATA 57, D0, 18, DA, 09, 10, 02, C8, 03, 00, 0C, 06, 07, 38, 0D, 04  
200 DATA FF, FF, @
```

Next issue I will cover how to read command line variables and start on the new commands.